NST DVA · CAPSTONE 2 · TRANSPORTATION & PUBLIC SAFETY

Severity Reality Check

Disentangling true accident severity from congestion, time-of-day, and location bias

ANALYZING 95K US ROAD ACCIDENTS · 2016–2023

Newton School of Technology

Team: Shivam Mittal · Satyam · Members 3–7

Mentor: <fill> · Section: <fill> · Team ID: <fill>

GitHub: github.com/<team-handle>

Tableau: public.tableau.com/<dashboard>

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Public severity data is widely used — but it measures the wrong thing.

WHAT'S WRONG

- The most common severity field measures traffic-flow impact (short delay → long delay), not crash physical severity.
- A fender-bender at rush hour can register Severity 4. A fatal crash on a quiet road can register Severity 1.
- Insurers, DOTs, and logistics planners use this field directly in pricing models, capital plans, and routing — silently absorbing the bias.

OUR CORE BUSINESS QUESTION

How much of what's labeled "severe" in public US accident data is genuinely severe — and how should insurers, DOTs, and logistics planners adjust their decisions to use a bias-corrected signal?

→ *Decision: down-weight raw severity in pricing, capital allocation, and corridor routing.*

From 7.7M raw rows to a 95K balanced sample

ORIGINAL ROWS

7.7M
raw Kaggle dataset

WORKING SAMPLE

95,607
balanced across Sev 1–4

COLUMNS

49
incl. 13 derived features

STATES COVERED

49
6,800+ cities

CLEANING PIPELINE (notebooks/02_cleaning.ipynb)

- Drop high-null + zero-variance columns (End_Lat/Lng, Country, Turning_Loop)
- Validate geographic bounds (lat 24–50, lng −125 to −66)
- Cap Distance_mi outliers at 99th percentile
- Context-aware imputation (median by State + Month + Hour)
- Cast booleans to int for Tableau

KEY DERIVED FEATURES

- Severity_Label · Weather_Category · State_Region
- Time_of_Day · Season · Is_Rush_Hour · Is_Weekend
- Duration_min · Road_Feature_Count
- Is True Severe = Sev=4 AND Distance ≥ 0.5 mi

15 decision-relevant KPIs across 3 dashboards

D1 — Severity Reality Check

the hook

- Total Accidents
- Nominal Severe (Sev 4)
- True Severe (Sev 4 + Dist ≥ 0.5 mi)
- True Severe % ← headline
- Avg Duration of Severe

D2 — When & Where

the diagnosis

- Worst Season
- Peak Risk Hour (12 AM, not rush hour)
- Top State by True Severe (PA, not CA)
- Weekend vs Weekday Δ
- Cities Tracked

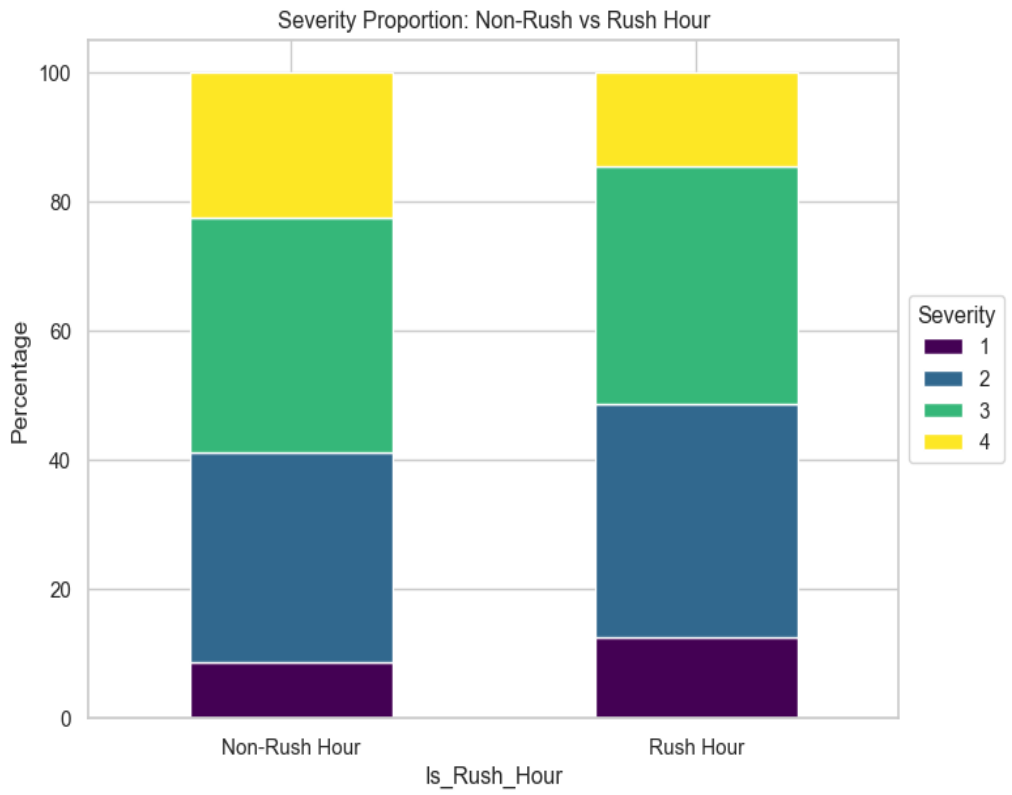
D3 — Conditions

the cause

- Riskiest Weather
- Junction Lift (+0.2–0.5 pts)
- Signal Drop (≈0.4 pts)
- Avg Visibility During Severe
- Night Severe Share

Five observations that reframe the data

- Severity-2 dominance dropped after balanced sampling — every tier now has enough N for cross-tier comparison.
- Volume peaks at 5 PM rush hour — but per-accident severity is barely different between rush and non-rush.
- Weekday volume rises Tue–Wed, falls weekend. Severity does not follow the same curve.
- Top-volume states: CA, FL, TX. Top True-Severe state: PA — geography differs once bias is corrected.
- Most accidents happen in fair weather. Severe accidents are no exception.



Severity proportions: Rush Hour vs Non-Rush — barely shift

Statistical tests aligned with the bias hypothesis

Test 1 — Congestion Bias

Method: Chi-square (Rush Hour × Severity)

Significant by N, but proportions barely shift. Rush-hour volume is NOT the dominant severity driver.

Test 2 — Duration × Severity

Method: Kruskal-Wallis H (non-parametric)

Median duration rises monotonically: Sev 1 ≈ 30 min → Sev 4 ≈ 210 min. Validates True-Severe definition.

Test 3 — Weather × Severity

Method: Chi-square (top 5 weather × Sev)

Statistically significant but small effect size. Weather is NOT the primary driver of severity.

Test 4 — Location Bias

Method: State × Severity cross-tab

PA, WY, MT, NY over-represented in Sev 3+4. Reporting bias is real and must be corrected.

Net finding: Severity in this dataset is partly real, partly noise. The True-Severe filter (Sev 4 + Dist ≥ 0.5 mi) cleanly separates them.

Three dashboards, one narrative: What → When/Where → Why

DASHBOARD 1
Severity Reality Check

KPI STRIP

Total · Nominal Severe · True Severe · True Severe % · Avg Severe Duration

CHARTS

Severity Donut · Nominal vs True Severe · Rush Hour Comparison · Hourly Severity Trend

DASHBOARD 2
When & Where

KPI STRIP

Worst Season · Peak Risk Hour · Top State by True Severe · Weekend vs Weekday Δ · Cities Tracked

CHARTS

Day × Hour Heatmap · Top 12 States · Time-of-Day Severity Bar · Region Comparison

DASHBOARD 3
Conditions

KPI STRIP

Riskiest Weather · Junction Lift · Signal Drop · Avg Visibility During Severe · Night Severe Share

CHARTS

Weather × Severity · Visibility Bucket × Severity · Road Feature Lifts · Temp × Sev by Season

6+ interactive filters across dashboards: Severity · State · Year · Time-of-Day · Season · Weather · Visibility

Five actions, each tied to an insight

#	STAKEHOLDER	RECOMMENDATION	EXPECTED IMPACT
1	**Insurer**	Discount publicly-reported Severity 4 by ~50%; switch to True-Severe-based pricing input.	*Mis-priced urban-policy margin recovered; more accurate risk-tier pricing.*
2	**State DOT**	Prioritize signal installations at top-N junction hotspots (from D3).	*≈ 0.4 pts severity reduction per retrofit corridor; measurable per-signal ROI.*
3	**Logistics fleet**	Re-rank corridors by True-Severe Index; shift dispatch off PA + winter night windows.	*Lower expected claim frequency and severity per million route-miles.*
4	**Infrastructure planner**	Rebalance away from visibility-only spend; invest in junction redesigns + signal coverage.	*2–3× improvement in $/severity-point reduction vs visibility-only.*
5	**Federal grants**	Reweight per-state safety allocation using True-Severe counts, not raw volume.	*Capital tracks real severity outcomes, not reporting frequency.*

Why this work pays off — directional estimates

INSURER · REC #1

~50%

of Severity-4 in pricing models is over-weighted; bias-correction unlocks ~$100M risk-adj. premium reallocation per $10B book.

STATE DOT · REC #2

–0.4 pts

average severity reduction at junctions retrofitted with traffic signals — quantifiable per-installation ROI.

FLEETS · REC #3

5–10%

expected reduction in severe-corridor exposure when routing on True-Severe Index instead of raw volume.

WHY ACT NOW

- Real-time accident APIs increasingly feed pricing engines — bias compounds quickly.
- Post-COVID vehicle-miles-traveled have rebounded; more decisions flow through the same biased input.
- Signal Drop is one of the most cost-effective infrastructure levers — and the data quantifies it.

Honesty about what this analysis can and cannot say

LIMITATIONS

- Severity is a flow-impact label, not a crash-injury label — True Severe is a proxy.
- 0.5-mi threshold is defensible but not absolute; sensitivity tested at 0.25 and 1.0 mi.
- All findings are correlational, not causal.
- API coverage varies by state; 2020+ volume spike is partly an artifact.
- No injury / fatality outcomes available in this dataset.
- Balanced sample changes base rates — most KPIs use ratios to control.

NEXT STEPS

- Join FARS / state crash records to validate True-Severe against real injury data.
- Add quasi-experimental causal layer for Signal Drop estimate.
- Wrap corrected severity index as a real-time API for fleets and insurers.
- Train a gradient-boosted classifier on incoming reports (predict True Severe at intake).
- Geospatial drill-down map (Mapbox / Leaflet) for DOT use.
- Attach cost-benefit modelling using FHWA retrofit cost data.

Built by Team — every member contributed

TEAM ROLES

Project Lead **Shivam Mittal**	Data Lead **Satyam**	ETL Lead **<Member 3>**	Analysis Lead **<Member 4>**
Visualization Lead **<Member 5>**	Strategy Lead **<Member 6>**	PPT & Quality Lead **<Member 7>**	

ARTEFACTS • github + tableau

- **GitHub Repo:** github.com/<team-handle> (5 notebooks · ETL pipeline · 3 dashboards · report + deck)
- **Tableau Public:** public.tableau.com/<dashboard> (3 views · 30+ worksheets · 6+ filters)
- **Project Report:** reports/project_report.pdf (18 sections, ~22 pages)
- **This Deck:** reports/presentation.pdf (11 slides)